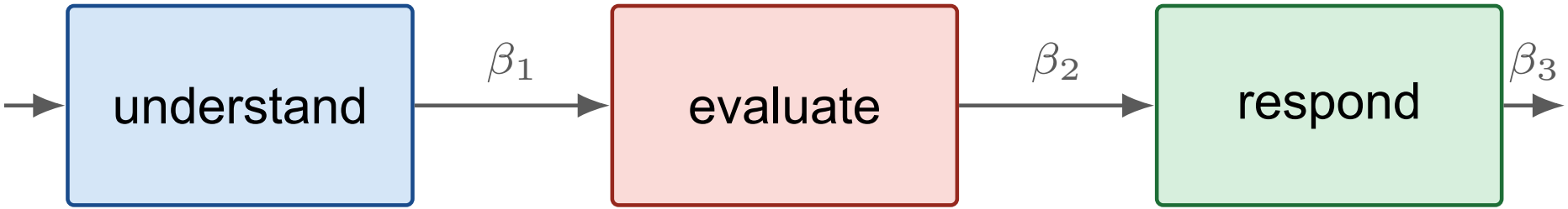

$$\beta_{\text{total}} = \beta_1 \cdot \beta_2 \cdot \beta_3$$

Gain multiplies along the chain. Flow does not — each stage draws its own current from its own local drive and resistance.